

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 2: Crew Composition & Flight Duty Limitations (FTL)

Syllabus Source: CAE Oxford (pp. 37-76) | Pass Target: >= 90% AviationExam

1. EASA FLIGHT AND DUTY TIME LIMITATIONS (FTL)

EASA Subpart-FTL (ORO.FTL) establishes mandatory limits to prevent crew fatigue from compromising safety:

- Flight Duty Period (FDP): Time from when a crew member is required to report for duty (typically 45 to 60 min before departure) until engine shutdown at the end of the last sector.
- Maximum Daily FDP: Ranges from 9 to 13 hours, depending on report time (circadian rhythm) and number of flight sectors (e.g. 1 to 2 sectors reporting between 06:00 and 13:29 allows 13.0 hours FDP; reporting between 02:00 and 04:59 drops to 9.0 hours!).
- Minimum Rest Period: Mandatory rest before duty must be AT LEAST 12 HOURS, or equal to the duration of the preceding duty period (whichever is GREATER!). If away from home base: at least 10 hours including an 8-hour sleep opportunity.

2. CUMULATIVE DUTY & FLIGHT TIME CAPS

- Maximum Cumulative Duty Hours: 60 duty hours in 7 consecutive days; 110 duty hours in 14 consecutive days; 190 duty hours in 28 consecutive days.
- Maximum Cumulative Flight Hours (Block Time): 100 flight hours in any 28 consecutive days; 900 flight hours in a calendar year; 1,000 flight hours in any 12 consecutive months.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Maximum Daily FDP Calculation

Problem: An acclimatized 2-pilot crew reports at 07:00 local for 4 sectors. Find the maximum legal Flight Duty Period under EASA ORO.FTL. Step 1: Check report time in ORO.FTL table: 07:00 (Favorable bracket 06:00 to 13:29). Step 2: Check sector column for 4 sectors: - 1-2 sectors: Max FDP = 13h 00m. - 3 sectors: Max FDP = 12h 30m. - 4 sectors: Max FDP = 12h 00m! Rule of Thumb: Each additional sector beyond 2 reduces allowed FDP by 30 minutes.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Minimum Rest Period after Heavy Duty

SCENARIO (EASA Flight Crew Rostering Drill): A commercial airline crew reports for duty at home base at 06:00 local time: - They operate a multi-sector European schedule - The duty finishes at 20:30 local time (Total duty period = 14.5 hours) QUESTION: What is the MINIMUM REST PERIOD required before the crew can report for their next duty? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the EASA Home Base Rest Rule: - Rule: Minimum rest at home base = The GREATER of: * (A) 12 hours standard baseline, OR * (B) The exact duration of the preceding duty period! Step 2: Compare the two values: - Standard Baseline = 12.0 hours - Preceding Duty Period = 14.5 hours - The greater value is 14.5 hours! Step 3: Calculate the Earliest Allowable Next Report Time: - Duty finished at 20:30. - Add 14.5 hours of rest: - 20:30 + 14h 30m = 11:00 the following morning! FINAL ANSWER: Minimum mandatory rest period is 14.5 hours (14 hours 30 minutes). Earliest next report is 11:00.

[!] EXAM TRAP (AVIATIONEXAM) - The Rest Away from Base vs Home Base Trap

- At HOME BASE: Rest = Greater of 12 hours OR length of duty. - AWAY FROM BASE: Rest = Greater of 10 hours OR length of duty (must include an 8-hour sleep opportunity). - Never assign 10 hours rest at home base; that is an illegal roster violation!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA FTL regulations, what is the maximum cumulative FLIGHT TIME (block hours) permitted in any 28 consecutive days?

- [A]: 60 hours
- [B]: 100 FLIGHT HOURS
- [C]: 110 hours
- [D]: 900 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. EASA ORO.FTL.210 sets maximum cumulative flight time at 100 hours in 28 consecutive days, 900 hours in a calendar year, and 1,000 hours in 12 months.

Q2: What is the maximum cumulative DUTY TIME permitted in any 7 consecutive days?

- [A]: 40 hours
- [B]: 60 DUTY HOURS
- [C]: 70 hours
- [D]: 100 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Total duty periods must not exceed 60 hours in 7 consecutive days, 110 hours in 14 days, and 190 hours in 28 days.

Q3: What is the minimum required rest period at home base following a duty period of 10 hours?

- [A]: 10 hours
- [B]: AT LEAST 12 HOURS (the baseline minimum)
- [C]: 8 hours
- [D]: 24 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At home base, rest must be at least 12 hours, or the length of the preceding duty if longer. Since $10h < 12h$, the 12-hour minimum applies.